BIOLOGY FOR ENGINEERS (BS-BIO101 / BS-BIO201)

Course Code: BS-BIO101 / BS-BIO201 | 3 Credits | L-T-P: 3-0-0

Maulana Abul Kalam Azad University of Technology - Official Curriculum Scheme

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Unit 1: Biomolecules & Cell Architecture

Prokaryotic and eukaryotic cellular organization, structural hierarchy of carbohydrates, lipids, proteins and nucleic acids, biological membranes, ion channels, active and passive transport.

Unit 2: Enzyme Kinetics, Metabolism & Cellular Energetics

Enzyme classification, active site catalysis, Michaelis-Menten kinetic parameters, enzyme inhibition mechanisms, glycolysis, Krebs TCA cycle, oxidative phosphorylation and ATP generation.

Unit 3: Molecular Genetics, DNA Replication & Protein Synthesis

DNA double helix architecture, semi-conservative replication, transcription mechanisms, translation, genetic code, recombinant DNA technology, PCR, CRISPR-Cas9 genome editing.

Unit 4: Microbiology, Immunology & Biosensors

Microbial growth kinetics, industrial fermentation, innate and adaptive immune response, monoclonal antibodies, biosensors transduction mechanisms, enzyme-based glucose biosensors.

Unit 5: Bio-inspired Engineering, Bio-materials & Bio-computing

Biomimicry in engineering design, lotus effect self-cleaning, gecko adhesion synthetic setae, biocompatible polymers, tissue scaffolds, DNA computing algorithms.